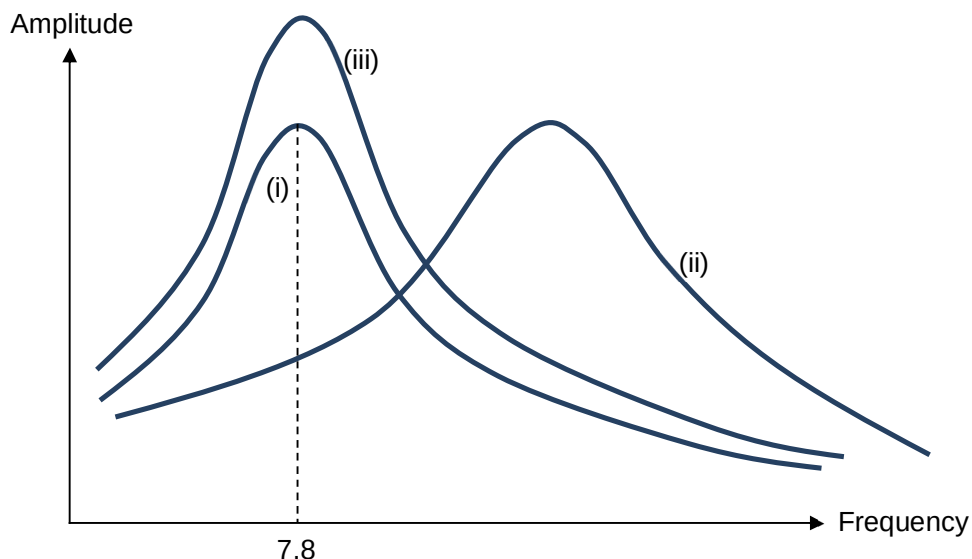


- 4 (a) (i)  $k = 3.0 \text{ N cm}^{-1} = 300 \text{ N m}^{-1}$  [1]  
 $\omega^2 = 2k/m = (2 \times 300) / 0.25$  [1]  
 $f = \omega / (2\pi) = 7.80 \text{ Hz}$
- (ii)  $\text{Max } v = \omega x_0$  [1]  
 $= 0.735 \text{ m s}^{-1}$  [1]

(b)



**Graph (i):** Shape [1]  
 Peak at 7.8 Hz (ecf for  $f$  from part a) [1]

**Graph (ii):** similar shape at higher  $f$  (same amplitude) [1]

**Graph (iii):** similar shape at same  $f$  (or slightly to the right) as graph (i) but higher peak [1]

(c) To maintain  $\omega$  when  $m$  changes to  $2m$ , need to increase  $k$  to  $2k$ . [1]

Add 2 similar springs, 1 in parallel to each spring. [1]

**OR**

Change to a (stiffer) spring with spring constant  $2k$ . [1]